# MILITARY SPECIFICATION SHEET

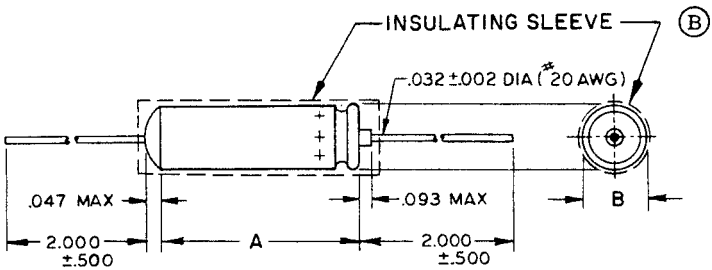
## CAPACITORS, FIXED, ELECTROLYTIC

## (DC, ALUMINUM, DRY ELECTROLYTE, POLARIZED)

## STYLES CE10 AND CE11

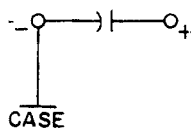
The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-62.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.




| INCHES | MM |
|---|---|
| .002 | .05 |
| .032 | .81 |
| .047 | 1.19 |
| .093 | 2.36 |
| .500 | 12.70 |
| 2.000 | 50.80 |

STYLES
CE10 – uninsulated
CE11 – insulated




UNINSULATED STYLE NOT FOR NEW DESIGN. FOR REPLACEMENT PURPOSES ONLY.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. These capacitors are not intended to be mounted by their leads. They shall be provided with a supplementary means of mounting such as a tangential bracket or wraparound bands. (Family 1.)

Ⓑ denotes changes

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FSC 5910

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TABLE I. Styles CE10 and CE11 capacitors.

| Type designation 1/ | Capacitance | DC rated voltage | DC leakage | Maximum impedance at 120 $+\frac{10}{-5}$ Hz at -40° $+\frac{0°}{-3°}$ C | Dissipation factor | Case dimensions 2/3/ A ±.031 (.79) | Case dimensions 2/3/ B ±.016 (.41) |
|---|---|---|---|---|---|---|---|
| | uf | volts | ua | ohms | percent | inches | inches |
| CE1-C560C | 56 | 5 | 2.0 | 178 | 20 | .688 (17.48) | .375 (9.53) |
| CE1-C750C | 75 | 5 | 2.0 | 133 | 20 | .812 (20.62) | .375 (9.53) |
| CE1-C101C | 100 | 5 | 2.0 | 100 | 20 | .938 (23.83) | .375 (9.53) |
| CE1-C181C | 180 | 5 | 2.0 | 55 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C271C | 270 | 5 | 2.0 | 37 | 20 | 1.500 (38.10) | .375 (9.53) |
| CE1-C150D | 15 | 10 | 2.0 | 600 | 20 | .688 (17.48) | .250 (6.35) |
| CE1-C390D | 39 | 10 | 2.0 | 230 | 20 | .688 (17.48) | .375 (9.53) |
| CE1-C560D | 56 | 10 | 2.1 | 160 | 20 | .812 (20.62) | .375 (9.53) |
| CE1-C101D | 100 | 10 | 2.3 | 90 | 20 | .938 (23.83) | .375 (9.53) |
| CE1-C151D | 150 | 10 | 2.7 | 60 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C221D | 220 | 10 | 3.2 | 41 | 20 | 1.500 (38.10) | .375 (9.53) |
| CE1-C100E | 10 | 15 | 2.0 | 800 | 20 | .688 (17.48) | .250 (6.35) |
| CE1-C270E | 27 | 15 | 2.0 | 296 | 20 | .688 (17.48) | .375 (9.53) |
| CE1-C560E | 56 | 15 | 2.3 | 143 | 20 | .812 (20.62) | .375 (9.53) |
| CE1-C750E | 75 | 15 | 2.9 | 107 | 20 | .938 (23.83) | .375 (9.53) |
| CE1-C101E | 100 | 15 | 3.3 | 80 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C151E | 150 | 15 | 4.1 | 53 | 20 | 1.500 (38.10) | .375 (9.53) |
| CE1-C5R6F | 5.6 | 25 | 2.3 | 1430 | 20 | .688 (17.48) | .250 (6.35) |
| CE1-C150F | 15 | 25 | 2.3 | 530 | 20 | .688 (17.48) | .375 (9.53) |
| CE1-C200F | 20 | 25 | 2.5 | 400 | 20 | .812 (20.62) | .375 (9.53) |
| CE1-C390F | 39 | 25 | 3.3 | 206 | 20 | .938 (23.83) | .375 (9.53) |
| CE1-C560F | 56 | 25 | 4.0 | 143 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C750F | 75 | 25 | 4.8 | 107 | 20 | 1.500 (38.10) | .375 (9.53) |

See footnotes at end of table.

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(B) TABLE I. Styles CE10 and CE11 capacitors. (Continued)

| Type designation 1/ | Capacitance | DC rated voltage | DC leakage | Maximum impedance at 120 $+10$ $-5$ Hz at $-40°$ $+0°$ $-3°$ C | Dissipation factor | Case dimensions 2/3/ A ±.031 (.79) | Case dimensions 2/3/ B ±.016 (.41) |
|---|---|---|---|---|---|---|---|
| | uf | volts | ua | ohms | percent | inches | inches |
| CE1-C1R0G | 1 | 50 | 3.3 | 7000 | 20 | .688 (17.48) | .250 (6.35) |
| CE1-C5R6G | 5.6 | 50 | 3.3 | 1250 | 20 | .688 (17.48) | .375 (9.53) |
| CE1-C100G | 10 | 50 | 3.5 | 700 | 20 | .812 (20.62) | .375 (9.53) |
| CE1-C150G | 15 | 50 | 4.4 | 470 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C220G | 22 | 50 | 5.0 | 320 | 20 | 1.250 (31.75) | .375 (9.53) |
| CE1-C330G | 33 | 50 | 6.0 | 210 | 20 | 1.500 (38.10) | .375 (9.53) |
| CE1-C8R2H | 8.2 | 100 | 8.0 | 850 | 15 | .938 (23.83) | .375 (9.53) |
| CE1-C100H | 10 | 100 | 8.0 | 700 | 15 | 1.250 (31.75) | .375 (9.53) |
| CE1-C180H | 18 | 100 | 9.4 | 390 | 15 | 1.500 (38.10) | .375 (9.53) |
| CE1-C1R0J | 1 | 150 | 10.0 | 7000 | 12 | .938 (23.83) | .375 (9.53) |
| CE1-C2R2J | 2.2 | 150 | 10.0 | 3200 | 12 | .938 (23.83) | .375 (9.53) |
| CE1-C5R6J | 5.6 | 150 | 10.0 | 1250 | 12 | .938 (23.83) | .375 (9.53) |
| CE1-C8R2J | 8.2 | 150 | 10.0 | 850 | 12 | 1.250 (31.75) | .375 (9.53) |
| CE1-C100J | 10 | 150 | 10.6 | 700 | 12 | 1.500 (38.10) | .375 (9.53) |

1/ Complete type designation shall include an additional digit to complete style.

2/ The dimensions shown are for style CE10; for style CE11, add .062 (1.57) ± .062 (1.57) to the nominal dimension.

3/ Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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REQUIREMENTS:

Dimensions and configuration:
See figure and table I.

Case type: Tubular, uninsulated (CE10) and insulated (CE11).

Capacitance value: See table I.

Capacitance tolerance: -10, +75 percent.

Operating temperature range:
-40° C to +85° C.

(B) Solderability: Method 208 of MIL-STD-202.

DC leakage: See table I.

Capacitance: Method 305 of MIL-STD-202.

Dissipation factor: See table I.

Terminal strength: Method 211 of MIL-STD-202, test conditions A and D.

Shock, medium impact: Method 205, test condition C, MIL-STD-202.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Temperature cycling and immersion:
Temperature cycling: Method 102, test condition D, MIL-STD-202.
Immersion: Method 104, test condition C, MIL-STD-202.
DWV-Insulating sleeves-Method 301 of MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial measurement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Moisture resistance: Method 106 of MIL-STD-202.
Insulation resistance (insulating sleeve) - Not less than 100 megohms.
DWV-Insulating sleeves-Method 301 of MIL-STD-202. 4,000 volts dc applied.
DCL - Not more than 150 percent of initial requirement.
Capacitance change - 5 percent.
DF - See table I.
Corrosion - Not more than 10 percent.

Stability at low and high temperature:
In accordance with MIL-C-62.
Step 1 (at +25° C):
DCL - See table I.
Capacitance - See table I.
DF - See table I.

Step 2 (at -40° C):
Impedance - See table I.

Step 3 (at +25° C):
DCL - See table I.
Capacitance - +5 percent of initial value.
DF - See table I.

Step 4 (at +85° C):
DCL - Not more than 10 times initial value, or meet initial 25° C requirement.
Capacitance - +20 percent of initial value.
DF - See table I.

Step 5 (at +25° C):
DCL - See table I.
Capacitance - ±5 percent of initial value.
DF - See table I.

Surge voltage: In accordance with MIL-C-62.
DCL - See table I.
Capacitance change - 6 percent.
DF - See table I.

Vent: Not applicable.

Life: Method 108, test condition F, MIL-STD-202.
DCL - See table I.
Capacitance change:
100 volts, dc rated, and less:
-20, +15 percent.
DC rated voltage above 100 volts:
±15 percent.
DF - Meet the initial requirement or not exceed 150 percent of the initial measurement.

Barometric pressure: Method 105 of MIL-STD-202.

Shelf life: In accordance with MIL-C-62.
Build factor: Not more than 4.0.
Shelf factor: Not more than 3.0.
DCL: Not more than 150 percent of initial value.
Capacitance change: 6 percent.
DF: Not more than 175 percent of initial requirement.

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*TABLE II. Supersession data.*

| Superseded insulated part number | Superseded uninsulated part number | Recommended replacement style CE10 or CE11 |
|---|---|---|
| 001 | 063 | CE1-C560C |
| 002 | 064 | CE1-C750C |
| 003 | 065 | CE1-C101C |
| 004 | 066 | CE1-C181C |
| 005 | 067 | CE1-C271C |
| 006 | 068 | CE1-C150D |
| 007 | 069 | CE1-C390D |
| 008 | 070 | CE1-C560D |
| 009 | 071 | CE1-C101D |
| 010 | 072 | CE1-C151D |
| 011 | 073 | CE1-C221D |
| 012 | 074 | CE1-C100E |
| 013 | 075 | CE1-C270E |
| 014 | 076 | CE1-C560E |
| 015 | 077 | CE1-C750E |
| 016 | 078 | CE1-C101E |
| 017 | 079 | CE1-C151E |
| 018 | 080 | CE1-C5R6F |
| 019 | 081 | CE1-C150F |
| 020 | 082 | CE1-C200F |
| 021 | 083 | CE1-C390F |
| 022 | 084 | CE1-C560F |
| 023 | 085 | CE1-C750F |
| 024 | 086 | CE1-C010G |
| 025 | 087 | CE1-C5R6G |
| 026 | 088 | CE1-C100G |
| 027 | 089 | CE1-C150G |
| 028 | 090 | CE1-C220G |
| 029 | 091 | CE1-C330G |
| 030 | 092 | CE1-C8R2H |
| 031 | 093 | CE1-C100H |
| 032 | 094 | CE1-C180H |
| 033 | 095 | CE1-C1R0J |
| 034 | 096 | CE1-C2R2J |
| 035 | 097 | CE1-C5R6J |
| 036 | 098 | CE1-C8R2J |
| 037 | 099 | CE1-C100J |

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Application data:
All capacitors shall withstand an rms ripple current at 120 Hz and 85° C as determined from the formula:

$$I_r = k\sqrt{C}$$

Where:

$I_r$ = rms ripple current in milliamps.
C = nominal capacitance in microfarads.
k = 10 for 0.25 inch dia cases.
= 20 for 0.375 inch dia cases.

Ⓑ Supersession data:
See table II.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - EC, AS, OS
Air Force - 11, 17, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

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